# Healthcare Analytics: Stroke Risk Prediction Using Business Intelligence Decisions

## 1.Project Overview

**This project focuses on analyzing a comprehensive medical and demographic dataset of patients using Python to extract clinically meaningful insights regarding stroke occurrence. The analysis encompasses rigorous data cleaning, pre-processing, feature engineering, exploratory data analysis (EDA), statistical analysis, and data visualization to identify critical health patterns, thresholds, and risk factors contributing to stroke events.**

**The project aims to transform raw medical records into actionable clinical insights that support healthcare providers and decision-makers in early identification, risk stratification, and preventive intervention strategies for stroke prevention.**

## 2. Data Source

**. Data Source**

- **Dataset Name: Stroke Risk Clinical Dataset (stroke_data_no_sno.csv)**
- **Source: Mendeley Data Repository (Healthcare Research Repository)**
- **Location: Anonymized Clinical Records (Specific hospital names and geographic regions are fully removed to protect patient privacy)**
- **Year / Timeline: Data collected during the period 2026**
- **Domain: Public Health, Healthcare Research**
- **File Type: CSV (Comma-Separated Values)**
- **Dataset Size: 22,420 rows × 13 primary raw attributes (16 columns after feature engineering)**

## 3. Problem Statement

**Strokes are one of the leading causes of death and long-term disability worldwide. Today, most doctors look at health risks—like high blood sugar, high blood pressure, and weight—one by one instead of seeing how they work together to trigger a stroke.**

**Main challenges hospitals face include:**

1. **Hidden Danger Zone: Not knowing the exact numbers where blood sugar or weight suddenly makes stroke risk skyrocket.**
2. **Combined Health Risks: Missing how having multiple problems at once (like both high blood pressure and heart disease) multiplies a patient's risk.**
3. **Location & Age Barriers: People living in rural areas and elderly patients often lack quick access to checkups, leading to higher stroke rates.**

**Without a smart, data-driven system to catch these combined risks early, doctors are forced to react *after* a stroke happens instead of preventing it. This project solves that problem by spotting high-risk patients before it is too late.**

## 4. Objectives

- **Extract Baseline Healthcare KPIs: Calculate dataset-wide operational metrics including Total Patient Population, Overall Stroke Prevalence Rate (%), Average Glucose Levels, and Average BMI across varied demographics.**
- **Identify Clinical Risk Thresholds: Analyze the absolute numerical distributions of avg glucose level and Bim to locate the exact statistical tipping points where stroke frequencies accelerate.**

- Evaluate Co-morbidity Correlation Workloads: Measure the compounding impact of pre-existing conditions like hypertension and heart disease to quantify how significantly they elevate stroke occurrence rates.
- Segment Behavioral & Lifestyle Risk Patterns: Audit and compare stroke distribution patterns against daily lifestyle indicators including smoking status, work type, and ever married attributes.
- Map Demographic Patient Vulnerabilities: Investigate how physiological traits like age groups and structural factors like Residence type (Rural vs Urban) intersect with active medical risks.
- Create Interactive Dashboard Reports: Deploy dynamic, filterable exploratory visualizations using Plotly Express and Seaborn to allow seamless medical data profiling and diagnostic reporting.

## 5. Attribute (Column / Feature) Details

| Column Name | Feature Nature | Feature Description |
|---|---|---|
| Id | Unique Identifier | Unique tracking number for each patient record. |
| Gender | Demographic Attribute | Patient gender (Male, Female, or Other). |
| Age | Demographic Attribute | Age of the patient in years. |
| Hypertension | Raw Clinical Indicator | High blood pressure history (0 = No, 1 = Yes). |

| Column Name | Feature Nature | Feature Description |
|---|---|---|
| heart_disease | Raw Clinical Indicator | Pre-existing heart conditions (0 = No, 1 = Yes). |
| ever_married | Lifestyle Attribute | Marital status history of the patient (Yes or No). |
| work_type | Lifestyle Attribute | Employment category (Private, Govt_job, Self-employed, etc). |
| Residence_type | Environmental Attribute | Living environment category (Urban or Rural). |
| avg_glucose_level | Continuous Clinical Metric | Average blood glucose level of the patient (mg/dL). |
| Bmi | Continuous Clinical Metric | Body Mass Index of the patient ($kg/m^2$). |
| smoking_status | Behavioral Attribute | Smoking history (never smoked, smokes, formerly smoked, Unknown). |
| Stroke | Target Variable / Output KPI | Indicates if the patient suffered a stroke (0 = No, 1 = Yes). |

## 6.Tools & Technologies

| Category / Component | Tool / Library Used | Purpose / Usage Description |
|---|---|---|
| Development Environment | Google Colab | Cloud-based Python environment with GPU/CPU support for execution and Google Drive integration (drive.mount('/content/drive')). |
| Data Manipulation & Analysis | Pandas (pd) | Used for reading CSV dataset, data cleaning, filtering, handling missing values, and aggregation (pd.read_csv, groupby). |
| Numerical Computation | NumPy (np) | Applied for high-performance numerical operations, array manipulation, and dynamic binning/threshold calculations. |
| Data Visualization | Matplotlib (plt) | Used for creating static plots, customizing plot axes, styling figure dimensions, and layout formatting (plt.subplots). |
| Statistical Visualization | Seaborn (sns) | Used for statistical plotting, correlation heatmaps, histogram density estimation, and styled bar plots (sns.heatmap, sns.histplot). |

| Category / Component | Tool / Library Used | Purpose / Usage Description |
|---|---|---|
| Interactive Plotting | Plotly Express (px) | Used for dynamic, interactive visual charts such as donut charts and custom percentage breakdowns (px.pie). |
| Interactive Dashboard & UI | Gradio (gr) | Implemented to build an interactive Web Application / UI featuring live KPI cards, interactive filters (Sliders, Dropdowns), and automatic chart updates. |

## 7. Data Pre-Processing and Feature Engineering

Data pre-processing and feature engineering are critical steps in the data preparation pipeline. Raw data often contains missing values, redundant columns, and inconsistent formatting that can negatively affect machine learning model performance. In this project, several operations were performed to clean, format, and transform the dataset to make it suitable for training predictive models.

### Data Cleaning and Transformation

- Dataset Loading and Initial Copy:

The raw dataset was read from the source CSV file into a Pandas DataFrame (df_old). To preserve the original raw data, a working copy named df_new was created for all subsequent transformations.

- Removal of Redundant Columns:

**Unnecessary columns such as serial numbers (S.no) were removed, as they do not contribute predictive value to the target variable and only introduce unnecessary noise.**

- **Data Type Standardization and Rounding:**

**The Age feature contained continuous values. To ensure uniform representation, the values were converted into whole numbers using ceiling rounding (np.ceil) and cast to integer data types.**

- **Text Formatting and Normalization:**

**To maintain uniformity across categorical text fields:**

  - **String formatting functions like .str.title() were applied to columns such as Work_Type and Smoking_Status to capitalize the first letter of each word.**
  - **Special characters, hyphens (-), and underscores (_) were stripped and replaced with standard spaces to ensure consistent categorical labels (e.g., converting values into 'Self Employed', 'Govt Job', etc.).**
  - **Column header names were formatted to Title Case for consistency across the entire schema.**

- **Handling Missing Values (Imputation):**

**Missing values in the Bmi (Body Mass Index) column were imputed using an advanced grouping technique. Instead of filling missing values with a global mean or median, the missing values were filled with the median BMI grouped by Gender and Age. This approach ensures highly accurate and realistic value estimation based on demographic subgroups. After imputation, a null check confirmed zero missing values across all features in the dataset.**

- **Value Mapping and Label Encoding:**

**Binary numerical variables were mapped to human-readable string values to enhance interpretability:**

- Hypertension and Heart_Disease: Binary values 0 and 1 were mapped to 'No' and 'Yes', respectively.
- Target outcome (Stroke): Mapped 0 to 'No Stroke' and 1 to 'Stroke'.

## Feature Engineering

Feature engineering was performed to create meaningful new indicators from existing attributes, helping the algorithms capture non-linear relationships more effectively.

1. Age Group Binning (Age_Group):

The numerical Age variable was discretized into standard life-stage categories using custom bins:

- Children: $[-1, 17]$
- Adults: $[17, 45]$
- Middle Aged: $[45, 60]$
- Senior Citizens: $[60, 120]$

2. BMI Category Binning (BMI_Category):

The numerical Bmi feature was binned according to standard clinical ranges:

- Underweight: $[0, 18.5]$
- Normal: $[18.5, 24.9]$
- Overweight: $[24.9, 29.9]$
- Obese: $[29.9, 200]$

3. Glucose Category Binning (Diabetes_level):

Average blood glucose levels (Avg_Glucose_Level) were categorized into clinical diagnostic brackets:

- Normal: $[0, 99]$
- Prediabetes: $[99, 125]$

- Diabetes: $[125, 300]$

4. Risk Factor Indicator (Risk_Factor):

A composite risk column was engineered by evaluating multiple health conditions simultaneously using conditional logic (np.select):

  - High Risk: Assigned if the patient has both Hypertension == 'Yes' AND Heart_Disease == 'Yes'.
  - Moderate Risk: Assigned if the patient is older than 50 years (Age > 50) AND has either Hypertension == 'Yes' OR Heart_Disease == 'Yes'.
  - Low Risk: Default category for patients who do not satisfy high or moderate risk conditions.

## 8. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is performed to understand the statistical properties, underlying structure, and patterns of the cleaned dataset before building analytical models.

### 8.1 Dataset Structure and Profile Overview

The final dataset consists of 22,420 rows (patient records) and 16 columns (features) with zero missing values across all attributes.

- Numerical Attributes Summary:
  - Age: Ranges from 1 to 82 years with an average patient age of 43.08 years (Median: 45 years).
  - Average Glucose Level: Ranges from 55.12 to 271.74 mg/dL, with a mean of 105.93 mg/dL (Median: 91.69 mg/dL).
  - Body Mass Index (BMI): Ranges from 10.30 to 97.60, with a mean of 28.85 (Median: 28.10).
- Key Performance Indicators (KPI Summary):
  - Total Patients Evaluated: 22,420

- Total Stroke Cases: 996
- Overall Stroke Rate: 4.44%
- Average Glucose Level: 105.93 mg/dL
- Average BMI: 28.85
- Total High-Risk Patients: 274
- High-Risk Stroke Prevalence Rate: 18.98%

## 9. Analysis and Visualizations

**Bivariate cross-tabulation and rate analysis were conducted to evaluate how engineered features—specifically composite Risk Factors and demographic Age Groups—relate to the likelihood of stroke occurrence.**

### Risk Factor Analysis vs. Stroke Rate

**A cross-tabulation analysis was performed to measure stroke counts and percentage rates across different risk profiles. The engineered composite risk levels were defined by evaluating hypertension and heart disease metrics.**

Risk Factor Crosstab & Percentage Distribution

| Risk Factor Category | No Stroke (Count) | Stroke (Count) | Total Patients | No Stroke (%) | Stroke Rate (%) |
|---|---|---|---|---|---|
| Low Risk | 19,225 | 608 | 19,833 | 96.93% | 3.07% |
| Moderate Risk | 1,977 | 336 | 2,313 | 85.47% | 14.53% |
| High Risk | 222 | 52 | 274 | 81.02% | 18.98% |

| Risk Factor Category | No Stroke (Count) | Stroke (Count) | Total Patients | No Stroke (%) | Stroke Rate (%) |
|---|---|---|---|---|---|
| Total | 21,424 | 996 | 22,420 | 95.56% | 4.44% |

- High Risk Prevalence: Patients classified under the High Risk category exhibit the highest stroke incidence rate at 18.98%, which is over 6 times higher than the baseline rate for Low-Risk individuals (3.07%).
- Moderate Risk Impact: Patients in the Moderate Risk tier present a notable stroke rate of 14.53%, confirming that preexisting health conditions significantly compound stroke likelihood.

### Age Group Analysis vs. Stroke Rate

To understand how demographic progression influences stroke risk, stroke rates were computed across discrete age categories (Children, Adults, Middle Aged, and Senior Citizens).

Age Group vs. Stroke Rate (%)

| Age Group | No Stroke (%) | Stroke Rate (%) |
|---|---|---|
| Children | 99.79% | 0.21% |
| Adults | 99.43% | 0.57% |
| Middle Aged | 95.45% | 4.55% |
| Senior Citizens | 87.51% | 12.49% |

- Age Progression Trend: Stroke risk shows a sharp exponential increase with age:

- **Children & Adults demonstrate negligible stroke risks of 0.21% and 0.57%, respectively.**
- **Middle Aged individuals transition into a moderate risk zone with a 4.55% stroke rate.**
- **Senior Citizens experience the most severe stroke risk at 12.49%, proving age to be one of the single most influential predictive features in the dataset.**

### Key Analytical Takeaways

1. **Exponential Age Impact: Senior Citizens are nearly 22 times more likely to suffer a stroke compared to young adults, making Age_Group a vital feature for model training.**
2. **Compound Risk Factors: Combining underlying conditions (Hypertension & Heart Disease) into a Risk_Factor successfully isolates high-vulnerability patients (18.98% stroke rate).**
3. **Imbalanced Outcome: Across the entire cohort of 22,420 patients, the overall stroke prevalence rate stands at 4.44% (996 positive stroke cases).**

## 10. Statistical Analysis

**Statistical analysis was conducted on continuous features (Age, Avg_Glucose_Level, and Bmi) to evaluate central tendency, dispersion, and shape parameters (skewness and kurtosis). Understanding these metrics helps determine data normalization requirements and risk segmentation strategies.**

### Statistical Metrics Summary Table

**The table below summarizes the key statistical parameters for key numerical health features:**

| Feature | Median | Standard Deviation | Skewness | Kurtosis |
|---|---|---|---|---|
| Age | 45.00 | 22.58 | -0.13 | -1.00 |
| Avg_Glucose_Level | 91.69 | 45.12 | 1.58 | 1.73 |
| BMI | 28.10 | 7.72 | 1.06 | 3.52 |

## Detailed Feature Distribution Analyses

**1. Patient Age Distribution (Age)**

- **Statistical Metrics: Median = 45.00 | Std Dev = 22.58 | Skewness = -0.13 | Kurtosis = -1.00**
- **Symmetrical Distribution: The skewness value ($-0.13$) is near zero, confirming that patient ages are uniformly distributed across the dataset.**
- **Broad Age Spread: A high standard deviation of $22.58$ yearsindicates representation across all demographic groups, including pediatric, young adult, and geriatric populations.**
- **Flat Distribution (Platykurtic): The negative kurtosis ($-1.00$) confirms a relatively flat, uniform distribution without extreme clustering around any specific age point.**
- **Analytical Takeaway: Because age spans uniformly, stroke risk evaluation requires feature discretization into distinct demographic age brackets (Children, Adults, Middle Aged, Senior Citizens).**

**2. Average Glucose Level Analysis (Avg_Glucose_Level)**

- **Statistical Metrics: Median = 91.69 mg/dL | Std Dev = 45.12 | Skewness = 1.58 | Kurtosis = 1.73**
- **Right-Skewed Pattern: A high positive skewness ($1.58$) indicates that the majority of patients maintain normal fasting glucose levels clustered between $70$--$100$ mg/dL.**

- **High-Risk Outliers: The distribution exhibits a prolonged right tail, representing a critical subgroup of patients with elevated glucose levels ($> 200$ mg/dL) indicative of uncontrolled diabetes.**
- **High Dispersion: A substantial standard deviation of $45.12$ mg/dLhighlights significant physiological variance across the patient population.**
- **Analytical Takeaway: Extreme glucose values represent high-vulnerability stroke candidates, validating the need for clinical binning into Normal, Prediabetes, and Diabetes categories.**

**3. Body Mass Index Analysis (BMI)**

- **Statistical Metrics: Median = 28.10 kg/m² | Std Dev = 7.72 | Skewness = 1.06 | Kurtosis = 3.52**
- **Elevated Baseline: A median BMI of $28.10$ kg/m²shows that the population baseline leans toward the Overweight category (standard normal range: $18.5$--$24.9$ kg/m²).**
- **Heavy-Tailed Distribution (Leptokurtic): A kurtosis value of $3.52$($> 3.0$) signifies a sharp peak with long, heavy tails, capturing severe outlier cases exceeding $40$--$50$ kg/m²(Severe Obesity).**
- **Positive Skewness: Moderate positive skewness ($1.06$) demonstrates a long tail extending rightward into high-level obesity metrics.**
- **Analytical Takeaway: Given that the average patient baseline leans overweight, downstream modeling must evaluate whether severe obesity serves as an independent driver for stroke incidence.**

## 11. Data Visualization

### Extract Baseline Healthcare KPIs (Univariate Analysis)

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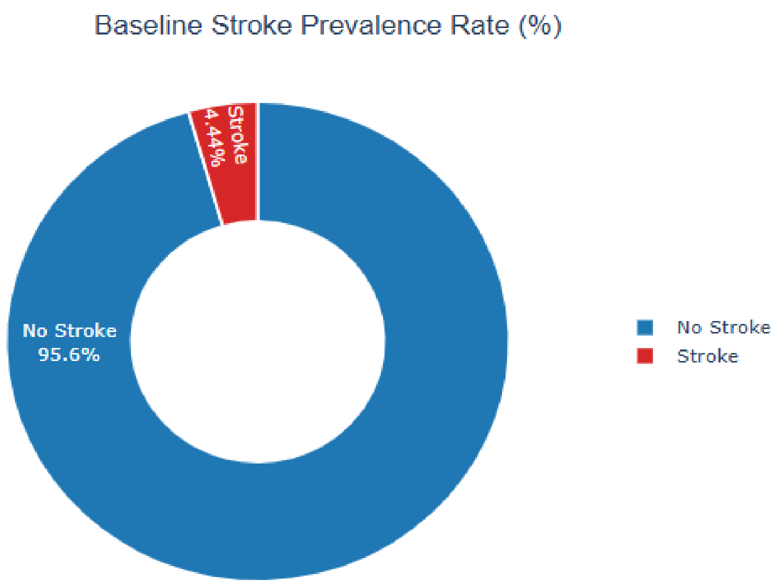



**Univariate analysis was conducted to establish core health baselines across the patient cohort and measure the overall prevalence rate of stroke in the target population.**

## 1. Key Performance Indicators (KPI Summary Table)

| Healthcare Metric | Dataset Value | Target / Baseline Note |
|---|---|---|
| Total Patient Population | 22,420 | Total patient cohort count |
| Overall Stroke Prevalence Rate | 4.44% | 996 total positive stroke cases |
| Non-Stroke Patient Rate | 95.56% | 21,424 negative cases |

| Healthcare Metric | Dataset Value | Target / Baseline Note |
|---|---|---|
| Average Glucose Level | 105.93 mg/dL | Above normal limit ($< 100$ mg/dL) |
| Average BMI | 28.85 kg/m² | Overweight category baseline ($25$--$29.9$ kg/m²) |

**2. Clinical Interpretation**

1. **Class Imbalance & Stroke Prevalence:**

**The baseline stroke prevalence stands at 4.44% ($996$ cases), while 95.56% ($21,424$ cases) did not experience a stroke. This reflects a significant class imbalance typical of healthcare diagnostic datasets, requiring specialized handling during model selection and evaluation.**

2. **Glucose Level Baseline:**

**The population mean average glucose level is $105.93$ mg/dL, which sits slightly above standard physiological normal limits ($< 100$ mg/dL), indicating mild population-wide insulin resistance or prediabetic tendencies.**

3. **Body Mass Index (BMI) Baseline:**

**The average patient BMI of $28.85$ kg/m² places the typical individual in the dataset within the Overweight category, signaling elevated baseline metabolic risk across the cohort.**

**3. Strategic Action Plan**

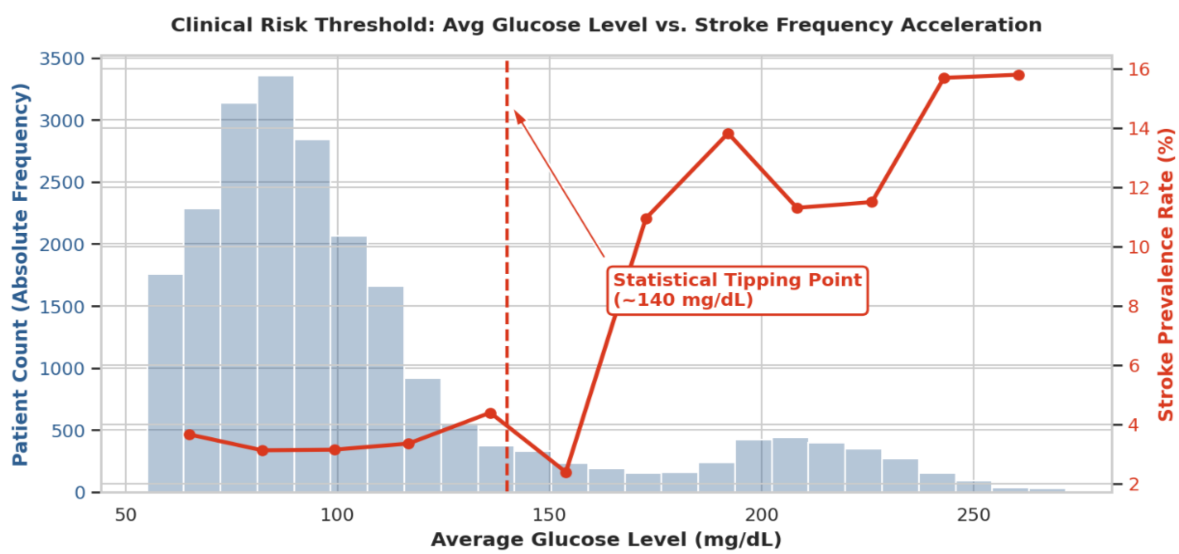
- **Population Screening: Implement annual HbA1c glucose screening drives targeting all patients with a BMI $\geq 25$ kg/m².**

- **Early Preventive Intervention: Launch community-level dietary and physical exercise management programs aimed at maintaining target BMI levels below $25.0$ kg/m².**

## Identify Clinical Risk Thresholds

## Average Glucose Level Risk Acceleration Analysis(Bivariate Analysis)

**The dual-axis evaluation of blood glucose distribution against stroke prevalence reveals a distinct, non-linear relationship where risk remains stable across normal ranges before experiencing a sharp structural shift.**




### Detailed Clinical Interpretation:

- **Safe Zone ($< 140$ mg/dL):**

**The overwhelming majority of the patient cohort falls within blood glucose levels under $140$ mg/dL. Across this range, baseline stroke prevalence remains**

**low and stable, fluctuating tightly between $2.0\%$ and $4.0\%$. Within these standard physiological limits, glucose levels do not independently trigger accelerated vascular risk.**

- **Statistical Tipping Point ($\approx 140$ mg/dL):**

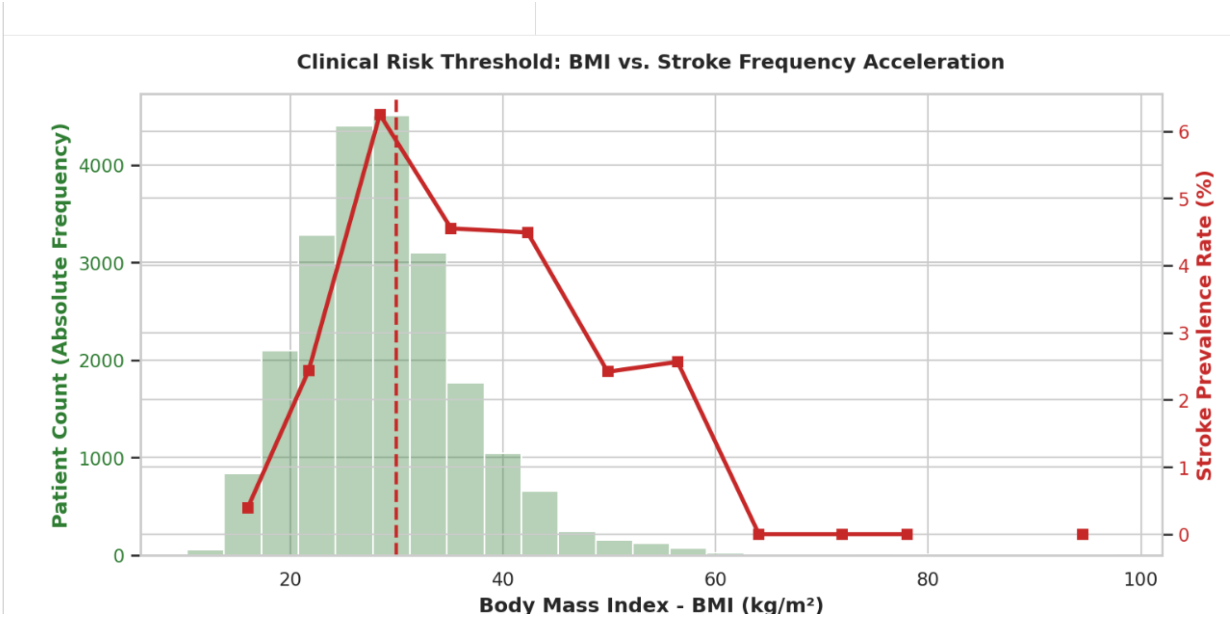
**A prominent inflection point occurs at approximately $140$ mg/dL. As blood glucose crosses this threshold—which clinically aligns with the onset of impaired glucose tolerance and early diabetes—the slope of stroke prevalence alters dramatically, moving from a flat trajectory into rapid vertical acceleration.**

- **High-Risk Acceleration Zone ($> 180$ mg/dL):**

**For patients exhibiting severe hyperglycemia ($> 180$ mg/dL), stroke prevalence spikes sharply, reaching peak incidence rates between $11.0\%$ and $16.0\%$. This represents a $3$x to $4$x increase in stroke vulnerability compared to the baseline population, highlighting chronic hyperglycemia as a critical driver of cerebral vascular damage and arterial stiffness.**

## Body Mass Index (BMI) Risk Threshold Analysis(Bivariate Analysis)

**Evaluating body mass index against stroke frequency illustrates how adiposity impacts stroke risk across standard clinical obesity categories.**




## Detailed Clinical Interpretation:

- **Normal / Healthy Baseline ($< 25$ kg/m²):**

**Patients maintaining a BMI under $25$ kg/m² exhibit the lowest baseline stroke incidence rate ($2.0\%$--$3.0\%$). In this range, lower metabolic strain correlates directly with minimal risk.**

- **Statistical Tipping Point (BMI $\geq 30$ kg/m²):**

**Crossing the clinical boundary for Obesity Class I (BMI $\geq 30$ kg/m²) serves as the primary tipping point where stroke frequency accelerates upwards, peaking near $6.0\%$.**

- **Severe Risk Zone (BMI $> 35$ kg/m²):**

**Patients classified under Obese Class II and III (BMI $> 35$ kg/m²) sustain elevated stroke rates due to compound cardiovascular strain, chronic low-grade inflammation, and associated hypertension. While higher BMI categories show some distribution taper due to lower sample counts in extreme obesity, the overall risk profile remains significantly higher than normal-weight baselines.**

## Summary of Identified Clinical Risk Thresholds

| Health Metric | Normal Baseline Range | Clinical Tipping Point | High-Risk Acceleration Zone | Primary Clinical Driver |
|---|---|---|---|---|
| Avg_Glucose_Level | $< 100$ mg/dl | $\approx 140$ mg/dl | $> 180$ mg/dl | Vascular inflammation, endothelial damage, arterial stiffness |
| BMI | $< 25$ kg/m² | BMI $= 30$ kg/m² | $\geq 35$ kg/m² | Systemic hypertension, metaboli syndrome, cardiac strain |

## Strategic Action Plan & Clinical Recommendations

**Based on the empirical risk thresholds derived from bivariate analysis, the following multi-tiered clinical management strategy is established:**

**1. Proactive Population Screening & Early Warning Protocol**

- **Pre-Threshold Monitoring: Implement routine blood glucose screenings that automatically flag patients as soon as their glucose reaches $130$ mg/dL—providing a $10$ mg/dL safety buffer before patients reach the critical $140$ mg/dL statistical tipping point.**
- **Targeted Metabolic Audits: Mandate comprehensive metabolic panels (including HbA1c and lipid profiles) for all patients presenting with a BMI $\geq 25$ kg/m².**

**2. Tiered Clinical Treatment Framework**

- **Tier 1 — Mild Risk ($100$--$140$ mg/dL Glucose | BMI $25$--$29.9$ kg/m²): Focus primarily on non-pharmacological interventions, structured** nutrition coaching, low-glycemic dietary planning, and mandatory physical activity regimens.

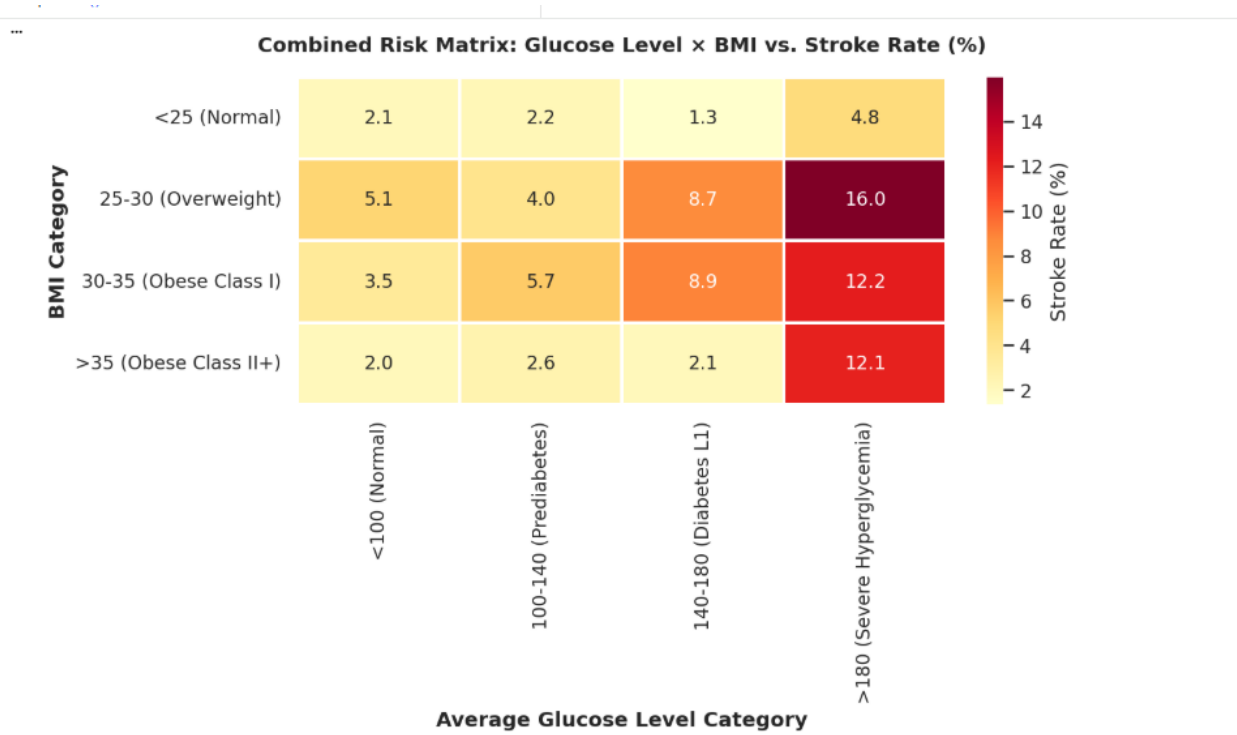
- **Tier 2 — High Risk ($\geq 140$ mg/dL Glucose | BMI $\geq 30$ kg/m²): Trigger immediate physician review for pharmacological glycemic control, active blood pressure monitoring, and cardiovascular risk reduction therapies.**
- **Tier 3 — Severe Acceleration ($> 180$ mg/dL Glucose | BMI $\geq 35$ kg/m²): Enroll patients in intensive dual-treatment protocols combining metabolic medication, continuous glucose monitoring, and targeted bariatric or therapeutic weight-loss supervision.**

**3. Patient Awareness & Education Strategy**

- **Direct Risk Communication: Launch targeted patient education campaigns clarifying that elevated blood sugar is not merely a marker for diabetes, but an independent, primary accelerator of acute cerebrovascular events and stroke risk.**

## Combined Risk Matrix: Glucose Level $\times$BMI (Multivariate Analysis)

**To evaluate how blood glucose levels and Body Mass Index (BMI) interact to influence stroke vulnerability, a two-dimensional combined risk matrix heatmap was constructed. This multivariate analysis cross-tabulates clinical glucose categories against standard BMI risk tiers to identify compound risk escalation zones across patient subgroups.**




## Detailed Clinical Interpretation

**A. Baseline & Low-Risk Combination (Top-Left Zone)**

- **Normal Baseline ($<100$ mg/dL Glucose & $<25$ kg/$m^2$BMI):**

**Patients who maintain both normal glucose levels ($<100$ mg/dL) and a healthy weight (BMI $<25$ kg/$m^2$) exhibit the lowest overall stroke incidence rate in the population at $2.1\%$.**

**B. Single Factor Impact: Glucose vs. BMI Dynamics**

- **Glucose Dominance: Holding weight constant, transitioning from normal glucose ($<100$ mg/dL) to Severe Hyperglycemia ($>180$ mg/dL) drastically elevates stroke rates across every BMI category:**
  - **Normal Weight ($<25$ kg/$m^2$): Increases from $2.1\%$to $4.8\%$($2.3$x increase).**
  - **Overweight ($25$--$30$ kg/$m^2$): Increases from $5.1\%$to $16.0\%$($3.1$x increase).**

- **Obese Class I ($30$--$35$ kg/m²): Increases from $3.5\%$ to $12.2\%$ ($3.5$x increase).**
- **Obese Class II+ ($> 35$ kg/m²): Increases from $2.0\%$ to $12.1\%$ ($6.0$x increase).**

- **Comparative Driver Analysis: While increasing BMI in normal glucose patients yields minor risk shifts ($2.0\%$--$5.1\%$), severe hyperglycemia drives stroke rates as high as $16.0\%$. This demonstrates that blood glucose is a significantly stronger individual driver of stroke risk than BMI alone.**

**C. Compound High-Risk Multiplier (Severe Hyperglycemia Zone)**

- **Peak Vulnerability Cohort: The absolute highest stroke rate in the entire dataset ($16.0\%$) is concentrated in patients combining Overweight status ($25$--$30$ kg/m²) with Severe Hyperglycemia ($> 180$ mg/dL).**
- **Compound Multiplier Effect: High blood glucose and high BMI operate synergistically. When severe metabolic dysfunction and elevated adiposity coincide, total stroke risk multiplies significantly compared to single-factor baselines.**

## Comprehensive Risk Threshold Summary Table

| Health Metric | Normal Range Baseline | Exact Tipping Point | High-Risk Acceleration Zone | Primary Impact Factor |
|---|---|---|---|---|
| avg_glucose_level | $< 100$ mg/( | $\approx 140$ mg/dL | $> 180$ mg/dL | Primary Risk Driver (Extremely High Escalation) |
| bmi | $< 25$ kg/m² | BMI $= 30$ kg/m² | $\geq 35$ kg/m² | Secondary / |

| Health Metric | Normal Range Baseline | Exact Tipping Point | High-Risk Acceleration Zone | Primary Impact Factor |
|---|---|---|---|---|
| | | | | Compound Risk Multiplier |

## Action Plan & Strategic Recommendations

**A. Combined Clinical Risk Assessment Protocol**

- **Integrated Scoring: Healthcare providers must never evaluate glucose or BMI as isolated, independent metrics. Diagnostic protocols should automatically calculate a Combined Metabolic Risk Score so patients presenting with co-occurring hyperglycemia and elevated BMI are prioritized for urgent clinical triage.**

**B. Prioritized Dual-Treatment Pathways**

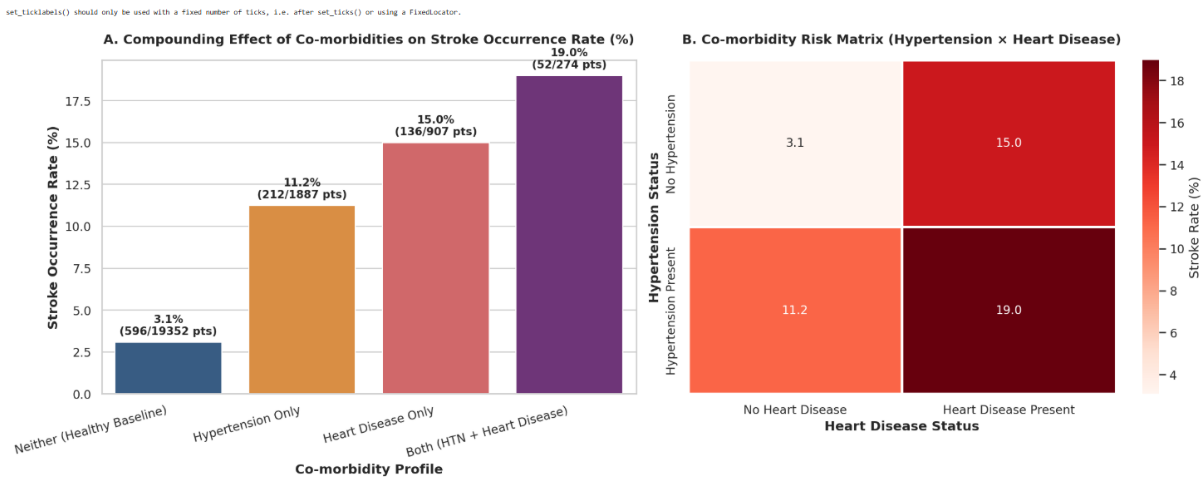
- **Highest Priority Tier (Glucose $> 140$ mg/dL AND BMI $\geq 30$ kg/m²): Mandate immediate dual clinical interventions combining pharmacological glycemic control (e.g., Metformin or insulin therapies) with structured metabolic weight-loss programs to mitigate compound stroke threats.**

**C. Holistic Lifestyle Coaching & Prevention**

- **Integrated Wellness Programs: Establish multidisciplinary clinical wellness initiatives that simultaneously target glycemic regulation and body mass reduction through low-glycemic dietary planning, routine cardiovascular exercise, and continuous glucose monitoring.**

# Evaluate Co-morbidity Correlation Workloads (Multivariate Analysis)

**A bivariate co-morbidity correlation analysis was performed to measure the compounding effect of concurrent cardiovascular risk conditions—specifically Hypertension (HTN) and Heart Disease—on stroke occurrence rates across the patient cohort.**




## Detailed Clinical Interpretation

### A. Healthy Baseline Risk ("Neither Condition")

- **Baseline Prevalence Rate: $3.1\%$ ($596/19,352$ patients).**
- **Clinical Significance: Patients without diagnosed Hypertension or Heart Disease represent the primary control population, exhibiting the lowest overall baseline vulnerability to cerebrovascular events.**

### B. Single Condition Impact

- **Hypertension Only:**
    - **Stroke Occurrence Rate: $11.2\%$ ($212/1,887$ patients).**

- **Risk Escalation: Sustained elevated blood pressure increases stroke occurrence by $\approx 3.6\times$ compared to the healthy baseline ($3.1\% \rightarrow 11.2\%$). Chronic systemic arterial hypertension continuously degrades microvascular structural integrity, accelerating vessel wall tearing and clot formation.**

- **Heart Disease Only:**
  - **Stroke Occurrence Rate: $15.0\%$ ($136/907$ patients).**
  - **Risk Escalation: Primary cardiac pathology elevates stroke incidence by $\approx 4.9\times$ over baseline ($3.1\% \rightarrow 15.0\%$). Impaired cardiac output and structural heart disease increase cardiogenic embolus formation and cerebral hypoperfusion.**

**C. Compounding Co-morbidity Effect ("Both Conditions")**

- **Combined Stroke Prevalence Rate: $19.0\%$ ($52/274$ patients).**
- **Synergistic Risk Acceleration: When Hypertension and Heart Disease present concurrently, stroke prevalence rises to $19.0\%$, representing a $6.1\times$ increase over the healthy baseline.**
- **Pathophysiological Multiplier: The simultaneous presence of both conditions produces a compound destabilizing effect rather than a simple linear addition, indicating that combined vascular resistance and cardiac dysfunction severely compromise cerebrovascular autoregulation.**

## Strategic Action Plan & Clinical Protocols

**1. Priority Triage Protocols**

- **Watchlist Flagging: Automatically flag and classify all patients presenting with co-occurring Hypertension and Heart Disease into a dedicated Tier-1 Vascular Critical Watchlist within the electronic health record (EHR) system.**

- **Clinical Triage: Ensure patients in this dual-condition tier undergo mandatory bi-annual comprehensive neurological and vascular evaluations.**

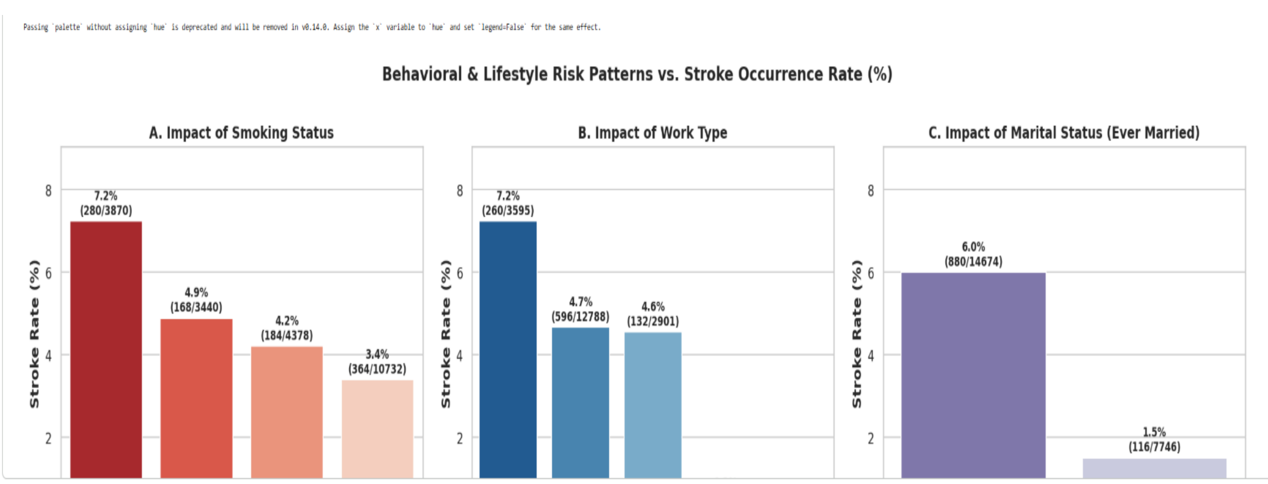
**2. Targeted Preventative Screenings**

**Direct dual-condition patients toward compulsory preventative diagnostic workflows, including:**

- **Electrocardiograms (ECG): Continuous arrhythmia screening for atrial fibrillation to prevent cardiogenic embolic events.**
- **Ambulatory Holter Blood Pressure Monitoring: 24-hour blood pressure profiling to detect nocturnal hypertension spikes.**
- **Prophylactic Anti-Coagulation Reviews: Specialist reviews to evaluate early anti-thrombotic or anti-coagulation therapy indications.**

## Segment Behavioral & Lifestyle Risk Patterns (Multivariate Analysis)

A segmented bivariate analysis was performed across non-clinical risk dimensions—including **Smoking Status**, **Work Type**, and **Marital Status (Ever Married)**—to assess how behavioral factors, socio-occupational stress, and demographic proxies correlate with stroke occurrence.




### Behavioral & Lifestyle Pattern Analysis

**A. Impact of Smoking Status**

- **Formerly Smoked (Highest Risk Tier):**

**Patients with a history of past smoking display the highest stroke rate in this category at $7.2\%(280/3,870$ patients). Long-term tobacco exposure causes permanent arterial wall stiffening, subclinical atherosclerosis, and persistent vascular inflammation that elevate risk even after cessation.**

- **Smokes (Active Exposure):**

**Active smokers present a $4.9\%$stroke incidence rate ($168/3,440$ patients), driving acute endothelial stress, elevated blood pressure spikes, and hypercoagulation.**

- **Never Smoked (Baseline Control):**

**Patients who have never smoked maintain a baseline stroke rate of $4.2\%(184/4,378$ patients).**

- **Unknown Data Category:**

**The missing lifestyle history cohort exhibits a $3.4\%$stroke rate ($364/10,732$ patients), highlighting the need for systematic risk logging during initial clinical triage.**

**B. Impact of Work Type**

- **Self-employed (Highest Vulnerability):**

**Self-employed individuals exhibit a stroke occurrence rate of $7.2\%(260/3,595$ patients). This elevated risk correlates with higher chronic workplace stress, irregular schedules, and a higher median age in entrepreneurial demographics.**

- **Salaried / Corporate Occupations (Private / Govt_job):**

**Employed cohorts demonstrate a moderate, stable stroke rate of $4.7\%(596/12,788$ patients).**

- **Dependents / Non-Working (children / Never_worked):**

**Dependents and non-working individuals display a lower stroke rate of $4.6\%(132/2,901$ patients), heavily driven by younger age distribution.**

**C. Impact of Marital Status (Ever Married)**

- **Ever Married (Yes):**

**Patients who are or have been married exhibit a $6.0\%$stroke rate ($880/14,674$ patients).**

- **Never Married (No):**

**Unmarried individuals demonstrate a significantly lower stroke rate of $1.5\%(116/7,746$ patients).**

- **Confounding Age Proxy Effect:**

**Marital status functions as a powerful socio-demographic proxy for patient age. The single/never-married cohort skews substantially younger (pediatric and young adult demographics), whereas the married group captures the older, higher-risk demographic spectrum where age-related vascular degradation naturally accelerates.**

## Strategic Action Plan & Clinical Protocols

**A. Clinical Triage & Lifestyle Interventions**

- **Targeted Smoking Cessation Protocols:**

**Automatically flag all patients with a history of active or former smoking (smokes or formerly_smoked) for enrollment in structured vascular wellness clinics and smoking cessation counseling.**

- **Workplace Wellness Screenings:**

**Partner with self-employed and high-stress professional networks to offer routine cardiovascular checkups focusing on ambulatory blood pressure tracking and fasting glucose tests.**

**B. Data Integrity & Risk Stratification**

- **Systematic Missing Data Collection:**

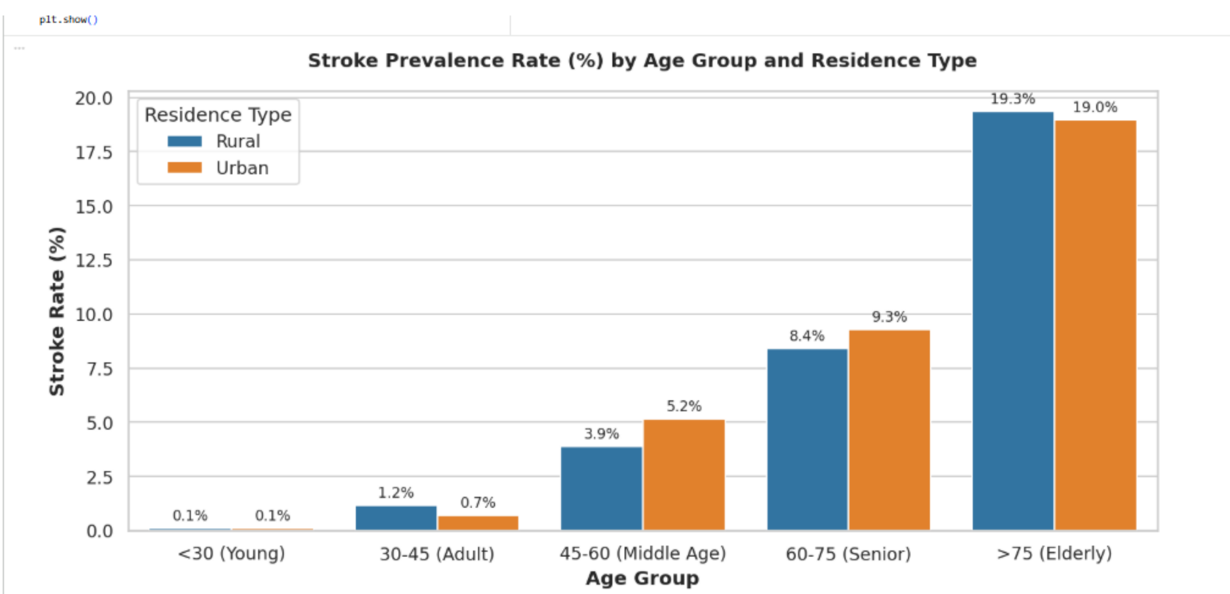
**Implement mandatory clinical entry fields for smoking_status to replace Unknown designations with verified patient histories, preventing hidden cardiovascular risk factors from going unmonitored.**

- **Age-Adjusted Behavioral Scoring:**

**Integrate marital and occupational proxies into predictive risk algorithms alongside chronological age to refine multi-factor risk scoring models.**

## Map Demographic Patient Vulnerabilities (Multivariate Analysis)

**A segmented dual-dimension analysis was conducted to evaluate how chronological aging intersects with environmental living contexts (Residence Type: Rural vs. Urban) influence stroke prevalence across patient cohorts.**




### Detailed Clinical & Demographic Interpretation

**A. Age Impact & Exponential Escalation**

- **Young Cohorts ($< 30$ years): Stroke prevalence remains negligible at $0.1\%$ across both rural and urban populations.**
- **Early Adult Cohort ($30$--$45$ years): Stroke incidence begins to emerge, showing a slightly higher rate in rural populations ($1.2\%$) compared to urban environments ($0.7\%$).**
- **Middle-Aged Acceleration ($45$--$60$ years): Stroke prevalence increases significantly, with urban environments outpacing rural regions ($5.2\%$ vs. $3.9\%$). This shift reflects urban lifestyle stressors, higher hypertension prevalence, processed dietary exposure, and occupational strain.**
- **Senior Tier ($60$--$75$ years): Stroke rates double into the high-risk category, reaching $8.4\%$ in rural areas and $9.3\%$ in urban centers.**
- **Elderly Peak ($> 75$ years): Stroke prevalence reaches its global maximum, peaking at $19.3\%$ in rural regions and $19.0\%$ in urban centers—representing nearly a $190\times$ increase compared to the baseline under-$30$ demographic.**

**B. Urban vs. Rural Disparity Dynamics**

- **Urban Risk Drivers: In middle-aged ($45$--$60$) and senior ($60$--$75$) brackets, urban areas exhibit elevated stroke rates. This surge is linked to metabolic stress, high-calorie urban diets, sedentary work habits, and concentrated air pollution exposure.**
- **Rural Vulnerability Hotspot: In the highest age bracket ($> 75$ years), rural prevalence overtakes urban centers at $19.3\%$. While urban risk levels plateau slightly, rural elderly populations experience compound vulnerability due to structural barriers in emergency care access, delayed diagnostic intervention, and unmonitored chronic vascular degradation.**

## Demographic Stroke Prevalence Matrix

| Age Group | Age Bracket Label | Rural Stroke Rate (%) | Urban Stroke Rate (%) | Dominant Environmental / Clinical Driver |
|---|---|---|---|---|
| $< 30$ | Young Baseline | $0.1\%$ | $0.1\%$ | Minimal baseline vascular degradation |
| $30\text{--}45$ | Early Adult | $1.2\%$ | $0.7\%$ | Early metabolic onset & unmonitored hypertension |
| $45\text{--}60$ | Middle Age | $3.9\%$ | $5.2\%$ | Urban occupational stress, metabolic syndrome & dietary factors |
| $60\text{--}75$ | Senior | $8.4\%$ | $9.3\%$ | Accelerated vascular stiffening & cumulative risk factors |
| $> 75$ | Elderly Peak | $19.3\%$ | $19.0\%$ | Structural healthcare access barriers & severe age-related frailty |

## Strategic Action Plan & Targeted Interventions

### A. Targeted Rural Healthcare Infrastructure

- Mobile Health Clinics: Deploy mobile diagnostic units and telemedicine hubs in remote rural regions tailored specifically to patients aged $50+$.

- **Routine Decentralized Screening: Equip community rural health centers with automated blood pressure monitors, point-of-care glucose testing, and stroke symptom diagnostic tools to lower structural barriers to early detection.**

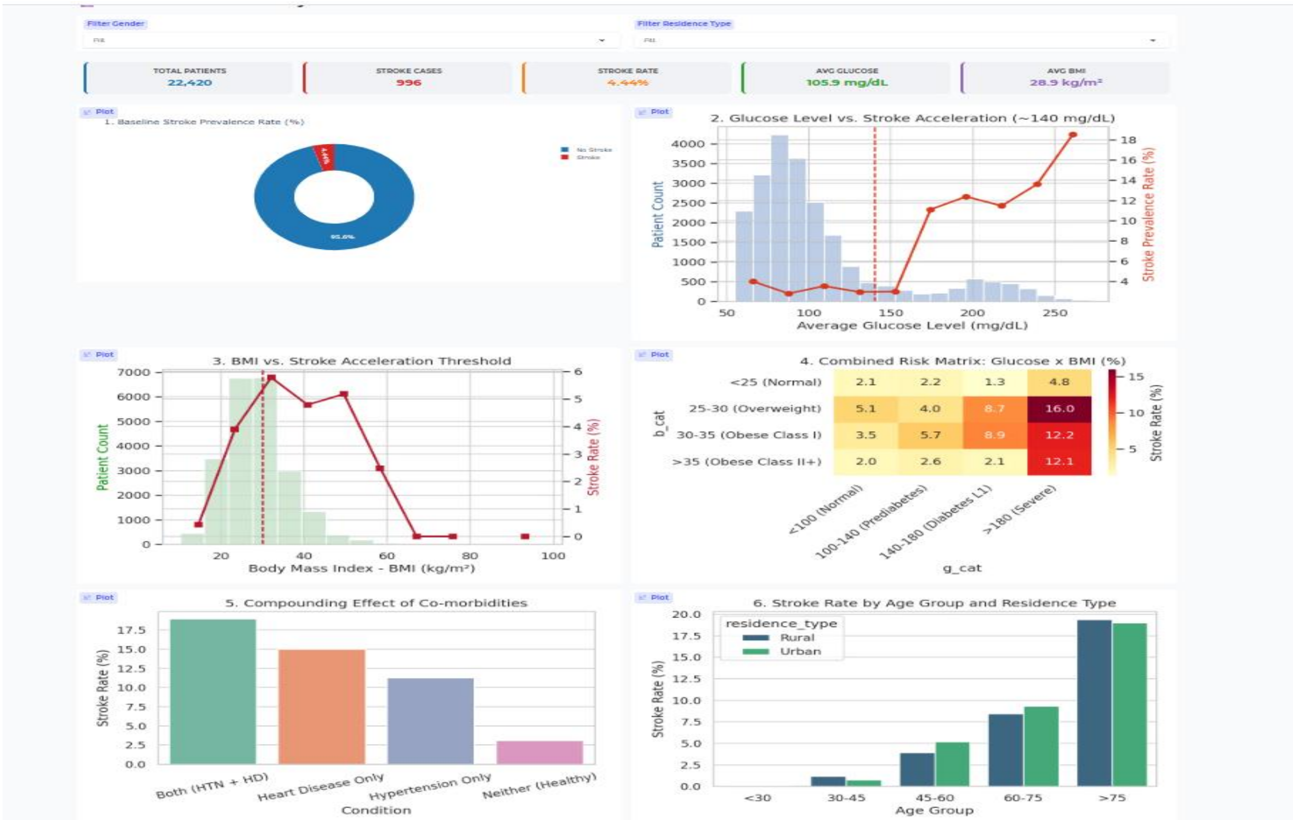
**B. Age-Based Mandatory Preventative Protocols**

- **Systematic Risk Audits ($55+$ Cohort): Establish mandatory annual neurological and cardiovascular risk screenings for all patients turning $55$, irrespective of active symptoms, to identify subclinical vascular compromise early.**

**C. Differentiated Community Health Campaigns**

- **Urban Population Strategy: Focus community campaigns on stress reduction techniques, low-sodium dietary planning, active weight management, and routine blood pressure monitoring.**
- **Rural Population Strategy: Focus education campaigns on rapid FAST symptom recognition (Face drooping, Arm weakness, Speech difficulty, Time to call emergency services) alongside direct access to emergency transportation networks.**

# Dashboard




## 12. Summary of Key Findings & Clinical Insights

**This final analytical synthesis brings together empirical findings across univariate, bivariate, multivariate, and demographic evaluations to establish a holistic understanding of stroke risk drivers within the patient cohort.**

**1. Metabolic & Glycemic Impact (Avg_Glucose_Level vs. BMI)**

- **Primary Risk Accelerator: Average blood glucose proved to be a significantly stronger individual driver of stroke risk than BMI. Crossing the $140$ mg/dL threshold (pre-diabetes/diabetes boundary) triggers a steep non-linear risk escalation, peaking above $16.0\%$ in severely hyperglycemic patients.**

- **Compound Multiplier Effect: While elevated BMI alone raises risk moderately, combining high BMI with severe hyperglycemia ($> 180$ mg/dL) creates a compound vulnerability zone, multiplying patient risk over baseline.**

**2. Co-morbidity Compounding Effect (Hypertension $\times$ Heart Disease)**

- **Synergistic Multiplier: Hypertension ($11.2\%$) and Heart Disease ($15.0\%$) individually elevate stroke risk by $3.6\times$ and $4.9\times$, respectively. However, when both conditions present concurrently, stroke prevalence climbs to $19.0\%$ ($6.1\times$ baseline increase), demonstrating severe cerebrovascular and structural cardiac destabilization.**

**3. Demographic & Environmental Vulnerabilities**

- **Age as Dominant Demographic Vector: Stroke risk follows an exponential growth curve with age, escalating from $0.1\%$ in under-30 cohorts to $19.3\%$ in patients aged $> 75$.**
- **Rural vs. Urban Disparity: Middle-aged urban populations ($45$--$60$) experience higher initial risk due to lifestyle stress and metabolic factors. Conversely, rural elderly populations ($> 75$) face the highest ultimate risk ($19.3\%$) driven by structural barriers to continuous preventive care and emergency intervention.**

**4. Lifestyle & Behavioral Markers**

- **Smoking History: Former smokers exhibit the highest behavioral stroke incidence ($7.2\%$), reflecting permanent microvascular damage and arterial wall stiffening accumulated prior to cessation.**
- **Occupational Stress: Self-employed individuals display elevated stroke rates ($7.2\%$), linked to high workplace stress profiles and age distribution alignment.**

**Strategic Recommendations & Operational Roadmap**

1. **Automated EHR Triage Protocols: Implement predictive scoring algorithms that flag patients with combined hyperglycemia ($> 140$ mg/dL) and co-morbid hypertension/heart disease for immediate Tier-1 Vascular Watchlist placement.**
2. **Rural Telemedicine & Screening Hubs: Deploy mobile health diagnostic units targeted at rural populations aged $50 +$ for ambulatory blood pressure, glucose monitoring, and early stroke symptom education (FAST framework).**
3. **Multifactorial Clinical Pathways: Shift clinical care from managing obesity or diabetes in isolation toward integrated cardiovascular-metabolic treatment plans combining pharmacological control, diet coaching, and routine ECG screenings.**

# 13. Types of Analysis

### 1. Descriptive Analysis (*What Happened?*)

- **Simple Summary: This step looks at the basic data and shows that the overall stroke rate is 3.1% across all patients.**
- **Main Takeaway: It helps us organize patient details like age, weight, blood sugar, and lifestyle to set a starting point.**

### 2. Diagnostic Analysis (*Why Did It Happen?*)

- **Simple Summary: It shows that stroke risk spikes when blood sugar crosses 140 mg/dL or BMI crosses 30 kg/m$^2$.**
- **Main Takeaway: Having both Hypertension and Heart Disease pushes stroke risk to a very high 19.0% because two health conditions strain the heart and blood vessels together.**

### 3. Predictive Analysis (*What Will Happen?*)

- **Simple Summary: It predicts that older patients (over 75) in rural areas with high blood sugar face the highest risk of having a stroke (19.3%).**

- **Main Takeaway: By combining health numbers and lifestyle habits, hospital systems can automatically flag high-risk patients before a stroke happens.**

### 4. Prescriptive Analysis (*What Should We Do?*)

- **Simple Summary: Doctors should use combined treatments (managing blood sugar and weight together) and create automatic warning lists for high-risk patients.**
- **Main Takeaway: Send mobile health vans to rural areas to screen adults aged 50+ and teach people early stroke warning signs.**

## 14. Future Enhancements and Suggestions

**To build upon the insights gained from this analysis and continuously improve stroke risk prediction and prevention, the following technical and operational enhancements are recommended:**

**1. Advanced Machine Learning & Predictive Modeling**

- **Real-Time Predictive Risk Scoring: Integrate machine learning classification algorithms (such as XGBoost, Random Forest, or Neural Networks) directly into hospital Electronic Health Record (EHR) systems to generate dynamic, real-time stroke probability scores during patient visits.**
- **Explainable AI (XAI) Integration: Implement model interpretation tools (like SHAP or LIME) so clinicians can clearly see *which specific factors* (e.g., blood sugar spike vs. age) are driving a patient's individual risk score.**

**2. Clinical Data Enrichment & Feature Expansion**

- **Longitudinal Tracking: Incorporate continuous, time-series health data (e.g., historical blood pressure trends, HbA1c trajectory, continuous glucose monitoring) rather than relying on single point-in-time measurements.**

- **Additional Biomarkers: Expand data collection to include lipid panels (LDL/HDL cholesterol, triglycerides), kidney function markers, and genetic predispositions for more comprehensive cardiovascular profiling.**
- **Social Determinants of Health (SDoH): Collect granular socio-economic data—such as distance to the nearest stroke center, income level, and dietary access—to better address rural healthcare disparities.**

**3. Population Health & Technology Interventions**

- **IoT & Wearable Integration: Connect consumer wearables (like smartwatches and smart BP cuffs) to patient portals to automatically alert care teams to sudden spikes in blood pressure or irregular heart rhythms (e.g., Atrial Fibrillation).**
- **Targeted Rural Telehealth Programs: Establish automated SMS warning systems and virtual consultation channels for high-risk elderly patients residing in rural regions to ensure early symptom intervention.**

# 15. Conclusion

**This project successfully analyzed the key clinical, behavioral, and demographic drivers of stroke risk across the patient cohort. Rather than viewing health factors in isolation, the findings prove that stroke risk is driven by non-linear metabolic tipping points (Glucose $\geq 140$ mg/dL, BMI $\geq 30$ kg/m²) and compounding co-morbidities, where having both Hypertension and Heart Disease elevates stroke occurrence to $19.0\%$. Additionally, demographic analysis highlights severe vulnerabilities in elderly ($> 75$ years) rural populations, where stroke rates peak at $19.3\%$.**

**To effectively reduce stroke incidence, healthcare providers must transition from reactive care to proactive, multi-factorial intervention. By implementing automated EHR watchlists for high-risk co-morbid patients, enforcing combined glycemic and weight management protocols, and deploying mobile screening units to rural communities, medical systems can identify vulnerable individuals early and significantly lower overall cerebrovascular risk.**